

SIDDHARTH BETALA

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RESEARCH INTERESTS

AI for Science, Graph Neural Networks, Generative Models, Agents, Evaluation, NLP, Multimodal Deep Learning

EDUCATION

Indian Institute of Technology Madras

Bachelor of Technology in Biological Engineering and Master of Technology in Data Science

July 2019 - July 2024

Chennai, India

◦ **CGPA: 9.02/10.00**

◦ **Thesis:** Enhancing Protein Fitness with Deep Learning: Sequence-Structure Fusion using Language Models and Graph Neural Networks for Function Prediction and Generative Sequence Design via Conditional Diffusion
Laxmi Ravi Prize for Best Thesis in Data Science Nomination

◦ **Scholarships:** MITACS Globalink Research Fellowship, University of Washington's IPD Summer Research Fellowship, Indian Academy of Sciences Research Fellowship, HTTA Scholarship

RESEARCH & PROFESSIONAL EXPERIENCE

Entalpic | Machine Learning Research Engineer 🌐

Supervisors: *Dr. Alexandre Duval, Dr. Victor Schmidt* | Works: [P.1, P.2]

January 2025 -

Paris, France

- **LeMat-Genbench:** Led the development of the first-ever unified evaluation benchmark for crystal generation models, standardizing **validity filtering**, **MLIP ensemble-based stability**, **uniqueness & novelty** metrics, and **distribution & diversity** analysis, released as an [open-source suite](#) with a [verified public leaderboard](#).
- **AgentAL:** Contributed to a DAG-orchestrated **active learning** library linking **GFlowNet sampling**, **MLIP-based evaluative scoring**, acquisition function-guided sample selection for DFT labelling, and iterative **GNN fine-tuning** to steer exploration toward high-reward catalysts, and identified **100+ candidates** for experimental validation.
- **LeMat-Synth:** Co-led the development of a [multi-modal extraction library](#) that combines **LLMs & VLMs** to convert unstructured materials-science literature into structured, ontology-aligned synthesis datasets; built large-scale pipelines spanning **OCR-based PDF parsing**, **agentic synthesis protocol extraction**, **figure segmentation & classification**, and **quantitative plot digitization**, enabling the creation of an **81k-paper corpus** and a [validated dataset](#) of **2.5k+ synthesis procedures** with expert-aligned **LLM-as-a-judge** evaluation.
- **Pathfinder:** Built a **hybrid retrieval system** for surfacing underrated materials-science patents, combining empirical comparison of patent-ranking algorithms, citation-graph expansion to a **40k-patent** corpus, and a novel **DSpy-based LLM evaluation agent** that generated scientist-style queries for automated top-k precision testing, culminating in **57 high-value patents delivered to a €250k client**.

Piramal Finance | Data Scientist 🌐

Rising Star in Business Intelligence Unit (BIU) Award, September 2024

July 2024 - December 2024

Mumbai, India

- Pioneered a division-wide feature selection pipeline using [feature-engine](#) and [BorutaPy](#), later **adopted as the standard** after presenting it at the BIU Townhall (Aug 2024).
- Developed a home-loan acquisition model that unlocked **Rs.4.8B in additional business** by reducing false-negative rejections through a **meta-learning ensemble of XGBoost and LightGBM**.
- Built a bank-statement narration tagging system by **LoRA-finetuning LLaMA2** on NER-augmented narrations and integrating it with a lightweight entity-extraction pipeline to robustly classify real-world transaction descriptions.
- Designed a two-track data science hackathon for talent recruitment from three premier institutes (IITs) and **mentored the top 10 finalists** through project development and evaluation.

MITACS Intern & Thesis Student 🌐

Supervisors: *Dr. Radhakrishnan Mahadevan (University of Toronto), Dr. Nirav Bhatt (IIT Madras)* | Works: [P.4, J.1]

May 2023 - June 2024

Toronto, Canada

- Developed a hybrid **ESM-GNN fusion model** integrating protein language model embeddings with structural graphs derived from OmegaFold using serial and parallel fusion strategies to enhance protein fitness prediction.
- Benchmarked convolutional and attention-based pooling architectures for sequence-to-function learning, revealing regime-dependent behavior between low-diversity & higher-diversity splits for mutation fitness.
- Demonstrated that **serial fusion substantially outperforms both sequence-only and structure-only baselines**, achieving state-of-the-art Spearman's ρ on GB1 and β -lactamase benchmarks.
- Designed a **diffusion-based** protein engineering pipeline using [EvoDiff](#) in-painting to generate function-enhanced variants, screened via the GNN predictor, identifying sequences surpassing wild-type fitness.
- Built large-scale processing pipelines: **ESM-2** embedding generation, **OmegaFold & ESMFOLD** structure prediction, graph featurization, and **pLDDT-based** filtering, for rapid model evaluation and candidate screening.
- Conducted systematic evaluation across sequence-to-function tasks from [FLIP](#) and [PEER](#) benchmarks, analyzing performance across data regimes, mutation complexity, and model initialization variance.

- Reproduced and rigorously benchmarked [ProteinMPNN](#) (GNN-based fixed-backbone design) and [MIF-ST](#) (masked inverse folding with sequence transfer) across multiple protein engineering tasks, validating key architectural claims from both papers.
- Demonstrated performance gains from **augmenting pre-training sets with AlphaFold2-predicted backbones**, improving sequence recovery relative to using PDB-only structures.
- Analyzed the effect of structure quality on inverse-folding performance, showing that **pretraining on AF2-predicted backbones with injected coordinate noise improves performance relative to unperturbed structures**, indicating that this featurization scheme mitigates potential model-specific structural biases.
- Built reproducible pipelines for PDB/AF2 preprocessing, graph featurization (inter-residue distances, dihedrals, planar angles), neighborhood construction, and masked-sequence training.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=POSTER, S=IN SUBMISSION, *=FIRST AUTHOR, #=SECOND AUTHOR

- [P.1*] "LeMat-GenBench: Bridging the gap between crystal generation and materials discovery." at AI for Accelerated Materials Design at NeurIPS 2025. [Spotlight](#) | [\[Paper\]](#) | [\[Code\]](#) | [\[Leaderboard\]](#)
- [P.2#] "LeMat-Synth: a multi-modal toolbox to curate broad synthesis procedure databases from scientific literature." In AI for Accelerated Materials Design at NeurIPS 2025. [\[Paper\]](#) | [\[Code\]](#) | [\[Dataset\]](#)
- [S.1*] "Out-of-Distribution performance as a proxy for explanation quality in graph neural networks." to be submitted to International Conference on Machine Learning (ICML) 2026.
- [S.2] "Revisiting and Revising RVL-CDIP: Quantifying Errors and Test-Train Overlap." under review at Conference of the European Chapter of the Association for Computational Linguistics (EACL) 2026.
- [C.1*] "A Picture is Worth a Thousand (Correct) Captions: A Vision-Guided Judge-Corrector System for Multimodal Machine Translation." System Description Paper at the Workshop on Asian Translation at IJCNLP-AAACL 2025. [\[Paper\]](#) | [\[Code\]](#)
- [C.2] "De-identification of sensitive personal data in datasets derived from IIT-CDIP." In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024. [\[Paper\]](#)
- [C.3*] "Leveraging LLM-Generated Contextual Conversations for Cross-Lingual Image Captioning." Systems Description Paper at the Ninth Conference on Machine Translation at EMNLP 2024. **Best Performance in Human Evaluation** | [\[Paper\]](#)
- [P.3*] "Out-of-Distribution performance as a proxy metric for graph neural network explainers in the absence of ground-truth explanations." In WiML Workshop at NeurIPS 2024.
- [P.4*] "Screening protein sequences generated via conditional diffusion for enhanced fitness using a GNN-based function predictor." In Machine Learning for Computational Biology (MLCB) 2024.
- [J.1#] "Advances in generative modeling methods and datasets to design novel enzymes for renewable chemicals and fuels." Current Opinion in Biotechnology 2023. [\[Paper\]](#)

STUDENT CONFERENCES

*=FIRST AUTHOR

- [1*] "Enhancing Protein Fitness: Sequence-Structure Fusion for Function Prediction and Generative Sequence Design" at WSAI Annual Research Showcase, Wadhwani School for Data Science and AI, IIT Madras, 2024. **Runner-up for Best Poster**
- [2*] "Sequence-informed structured GNNs to screen diffusion-generated proteins for enhanced function." Poster at Contemporary Perspectives in Computational Biology, Institute of Mathematical Sciences, 2024.
- [3] "Utilizing Whey Water to Produce Bacterial Cellulose-based Insulin Patch." with iGEM IIT Madras team (Team GEnoM) at National Bioengineering Conference 2022. **Unique Idea Award**
- [4] "Utilizing Whey Water to Produce Bacterial Cellulose-based Insulin Patch." with iGEM IIT Madras team (Team GEnoM) at Global Open Genetic Engineering Conference 2021. **Gold Medal for Best Computational and Overall Project** | [\[Paper\]](#)

PROJECTS

Out-of-Distribution (OoD) Performance as a Proxy for Explanation Quality in GNNs

September 2024 - March 2025

Supervisors: [Dr. Chirag Agarwal](#) (University of Virginia), [Dr. Guadalupe Gonzalez](#) (Genentech) | Works: [S.1, P.3]

- Introduced the **Explanation Generalization Score (EGS)**, a ground-truth-free metric that evaluates explainer quality by measuring model robustness improvements under distribution shifts.
- Conducted large-scale **empirical study with 4,800 trained models** across 7 explainers including GNNExplainer, GraphMask, Integrated Gradients, and gradient-based methods, demonstrating **up to 19% improvement in OoD performance** when explanation-guided GNNs are constrained with causal subgraphs.

- Created the **KINASE-I benchmark dataset** with domain-expert-curated ground truth molecular substructures, representing the **first dataset designed explicitly for evaluating explainers** in molecular property prediction.
- Established that **explainer reliability varies significantly between in-distribution and OoD settings**, with GraphMask maintaining consistency across distribution shifts while gradient-based methods show variable performance, providing practical guidance for deploying explanations in drug discovery applications.

LLM-Driven Cross-Lingual Image Captioning with Vision-Guided Data Correction

August 2024 - October 2025

English-to-Indic Multimodal Translation Shared Tasks [WMT 2024, WAT 2025] | Works: [C.1, C.3]



- Developed an automated **vision-guided judge-corrector pipeline** using multimodal LLMs (**GPT-4o-mini**, **Gemini 2.5 Flash Lite**) to evaluate and repair translation errors in low-resource multimodal datasets, processing **28,928 examples** per language and correcting **17.1%** of captions across four Indic languages.
- Designed an **DSPy-based routing system** that distinguishes between **visually ambiguous cases** (26.7%, corrected via VLM reasoning) and **pure mistranslations** (73.3%, retranslated via **IndicTrans2**), producing systematically cleaner, semantically grounded training data.
- Achieved consistent gains by **LoRA-finetuning IndicTrans2** on corrected data, including **+1.30 BLEU** (Eval) and **+0.70 BLEU** (Challenge) for English→Bengali, and **+0.60 BLEU** (Eval) for English→Odia.
- Introduced a **prompt-weighted captioning strategy** that balances the original English caption with LLM-generated contextual cues and securing **1st place in human evaluation** at WMT 2024 without any supervised fine-tuning.
- Performed the first large-scale **error taxonomy analysis** for this task, showing that Malayalam required the highest correction rate (24.0%) while Odia required the lowest (12.1%), and establishing that **visual ambiguity** and **poor translation quality** are the dominant error modes in Indic multimodal MT.

Hierarchical Loss Functions and Soft Labels for Mitigating Neural Collapse (NC)

Fall 2023

Course project under **Dr. Kaushik Mitra**, Project Grade: 10/10



- **Demonstrated NC prevalence** across multiple loss functions, including MSE and Cross Entropy on CIFAR-10 and ImageNet datasets using VGG and ResNet architectures, validating its universality.
- **Investigated hierarchical representation integration** to align similar classes more closely in learned feature space, examining whether hierarchical constraints counter typical NC properties in the terminal phase of training.
- Constructed **hierarchy tree for CIFAR-10** and implemented **soft label training** methodology to enforce class hierarchy, replacing one-hot encodings with distance-weighted probabilities based on hierarchical relationships.
- Designed a **custom hierarchical loss function** penalizing misclassifications based on class hierarchy distance rather than uniform penalties, aiming to reduce error severity by distinguishing similar from dissimilar confusions.

Addressing Data Quality Issues in RVL-CDIP: De-Identification and Dataset Revision

August 2023 - July 2024

Projects at **ML Collective** with **Stefan Larson** | Works: [S.2, C.2]

- Built an automated **PII de-identification pipeline** for IIT-CDIP-derived datasets, detecting and replacing sensitive entities (e.g., SSNs, phone numbers, addresses, religion, birthplace) using NER, regex matching, **Faker**-based synthetic substitution, and **Augraphy** / **Albumentations** augmentations.
- Developed a **filter-and-refine duplicate detection system** combining **CLIP** embeddings with **SuperPoint** / **LightGlue** feature matching and sequence similarity, identifying **35% test-train overlap**, with extreme leakage in categories such as specification (82.2%) and questionnaire (61.6%).
- Demonstrated the asymmetric impact of data-quality fixes: **error removal improves model accuracy** (e.g., LayoutLMv3: 94.30% → 97.32%), while **duplicate removal decreases scores** (3 points), revealing inflated benchmarks due to data leakage and showing zero-shot LLMs are most sensitive to error remediation.

Robotic Personal Assistant | Team Lead at Robotics Club, IIT Madras

2020-2022

Acquired by **American Express DART Lab, IIT Madras** at an undisclosed price for deployment in workplace monitoring systems

- Built a **vision pipeline** with OpenCV for face/emotion detection (Haar cascades) and YOLO-based object recognition using anchor boxes, non-maximum suppression, and Intersection over Union filtering.
- Implemented **ORB-SLAM2** with probabilistic deep-learning components for robust mapping and localization in dynamic, multi-room environments, visualized through RViz and tested in Gazebo.
- Integrated an **intent-based chatbot** and a lightweight **product recommendation system** using Amazon data scraping to enable interactive, context-aware user assistance.

TEACHING EXPERIENCE

Paid Teaching Assistantships | IIT Madras

Facilitated course logistics, grading, tutorials, auto-graded programming assignments, and student mentoring

- **BT3032 – Evolutionary Biology**, **Prof. Himanshu Sinha**

Fall 2023

- **BT1010 – Life Sciences**, **Prof. V Kesavan**

Spring 2024

Primary Instructor | CFI Summer School for Robotics, IIT Madras

Summer 2022

Led programming and machine learning modules for an intensive undergraduate robotics school catering to 130+ participants.

- Developed a comprehensive curriculum with Jupyter notebooks and project briefs spanning Python, OpenCV for Robotics, ROS, Linux CLI, and ML fundamentals.
- Led hands-on workshops and mentored projects on perception, controls, and deployment alongside 8 coordinators.

HACKATHONS

Temenos AI for Green Finance Hackathon

January 2024

Placed 3rd out of 340 teams across India & won prizes worth Rs.600,000

- Built a platform to evaluate green finance investment potential from project design documents (PDDs) using Donut embeddings for OCR-free multimodal parsing and semantic search.
- Deployed RAG-based interface with Claude on AWS Bedrock via Streamlit for carbon credit price projections.

Piramal Finance Data Science Hackathon

March 2024

Secured 1st place among 1000+ participants, winning prizes worth Rs. 100,000 and a full-time job offer.

- Developed an ensemble of classical ML models (XGBoost, LightGBM) and an attention-based architecture (TabTransformer) for a home loan acquisition scorecard, achieving an AUC-ROC of 0.88.

American Express Campus Challenge: Credit Card Default Prediction

November 2022

Achieved 2nd position and received a summer internship offer for the data scientist role



- Leveraged a bagging ensemble of an MLP, Catboost, XGBoost, and Histogram-based gradient boosting models after feature selection using variance inflation factor.

SENAI Lab Data Contest: Auto-grader Tool

May 2022

Placed 3rd with 82% semantic similarity



- Compared Doc2Vec, BERT, and RoBERTa on a [2500-question dataset](#) to score student answers v/s model solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB

ML/DL Libraries/Frameworks: PyTorch, PyG, torchgfn, sklearn, TensorFlow, wandb, MLFlow, statsmodels, nltk

Agentic Libraries/Frameworks: DSPy, LangChain, LangGraph, FAISS, Qdrant, CrewAI, PydanticAI

Other Technologies/Frameworks: Git, OpenCV, streamlit, SLAM, Gazebo, NumPy, pandas, matplotlib, Spark

COURSEWORK

Computer Science & Engineering: Deep Learning for Imaging, Pattern Recognition & Machine Learning, Database Management Systems, Data Structures & Algorithms, Big Data Lab, Data Analytics Lab

Applied CS & Data Science: Introduction to Data Analytics, Analysis & Interpretation of Biological Data, Computational Neuroscience, Biostatistics, Theoretical Biophysics, Computational Economics & Public Policy

Math: Linear Algebra, Probability & Statistics, Calculus, Series & Matrices, Mathematical Foundations of Data Science, Mathematical Modelling, Numerical Methods

ACADEMIC REVIEWING

AI for Accelerated Materials Design (AI4Mat) Workshop: NeurIPS 2025

Conference on Machine Translation (WMT): EMNLP 2025

XAI4Science Workshop: ICLR 2025, AAAI 2026

Machine Learning for Structural Biology (MLSB) Workshop: NeurIPS 2024, NeurIPS & EurIPS 2025

Machine Learning for Life and Material Science (ML4LMS) Workshop: ICML 2024

CO-CURRICULAR ACTIVITIES AND OTHER ACHIEVEMENTS

Open-Source and Open Science Communities	• HuggingFace Science: Contributing to DeepCritical, a research agent for structuring large-scale preclinical biology data into actionable knowledge for discovering cures. • ML Collective: Worked on NLP and GNN explainability projects, resulting in 1 EMNLP Main paper and 2 papers under review at ICLR and EACL 2026.
Technical Achievements	• Selected for the Amazon ML Summer School 2022. • 2-time Inter-IIT Technical Contingent member; ranked in the top 5 on both occasions.
Team iGEM/GenoM, IIT Madras	• Received a \$1500 Diversity in Science grant from The Biochemical Society, UK. • Produced a 6-part vaccine-awareness video series, translated and dubbed into 6 regional languages for wider accessibility.
Entrepreneurial Achievements	• Won a total of Rs. 500,000 from Entrepreneurship Cell IITM and Nirmaan IITM for developing a decentralized NAS-based storage system using personal devices.
Volunteering	• Mentored grade 9–12 underprivileged students in Vyasarpadi, Chennai, through classes and curated practice tests with the National Service Scheme. • Recorded and donated 5 audiobooks for visually challenged readers to Anna Centenary Library, Chennai, via the National Service Scheme. • Leading the development and deployment of an AI tutor in Hindi, Bhojpuri, and Maithili for high school students in Bihar with Turn the Bus.